

Marblehead → Provincetown — Overnight Race Weather

Friday 2026-07-17, 19:00 EDT start · ~03:00 Sat finish · ~40 nm SE across Massachusetts & Cape Cod Bays

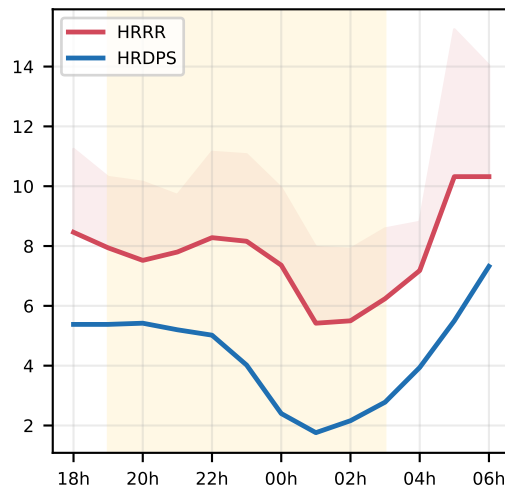
NOAA HRRR (3 km) and Environment Canada HRDPS (2.5 km) — two independent high-resolution models.

They agree on the overnight but split on the start — compared below and mapped on p.2. · 2026-07-16 22:09Z · ~24 h lead.

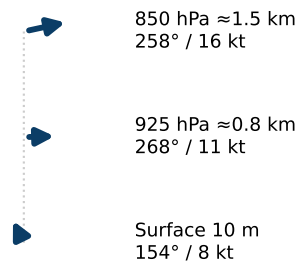
Route & overnight wind (HRRR mean arrows)



Course-mean wind (kt)



Wind shear — HRRR (mid-bay)



~100° veer, wind ~2x in 1 km.
Light SE surface under W-WNW ~16 kt.
925 hPa +1.2°C = inversion (decoupled).
→ full explainer + tactics on p.3

Overnight synopsis

Light, weak-gradient southerly night (flat ~1016 mb, no front). The models AGREE on the overnight: the wind fills S/SW 8-12 kt offshore, a starboard-tack reach that gradually frees toward Race Point, seas ≤2 ft. They SHARPLY DISAGREE on the start: NOAA HRRR has SE ~10 kt (a reach/beat down the rhumb) while Canada HRDPS shows a NW land-breeze ~7 kt (offshore), dying to near-calm by ~01:00 before the southerly fills — opposite first-leg angles. HRDPS has verified better locally this season, but at a post-sunset transition neither is reliable: sail the breeze that's actually on the water, then get offshore into the steadier southerly. Hour-by-hour, tactics, tide and hazards overleaf.

How the two models differ (this race)

Both are high-res mesoscale models and AGREE on the overnight (S/SW 8-12 kt reach to Race Point).

- HRRR (NOAA, 3 km): hourly, radar/satellite DA — strong on convection, gusts & fast change. Holds a decoupled SE surface here; gives a gust field.
 - HRDPS (Canada, 2.5 km): finer mesh; the better local performer this season. Mixes more → a NW land-breeze at the start; no separate gust field.
- They split most at the START (near-opposite dirs); they converge by the finish.

Conditions at a glance

Gusts — steady, modest

HRRR gust factor ~1.4-1.5, peaks ~15-16 kt over ~9 kt mean. CAPE 0 + only ~11-15 kt aloft → no convective or momentum-driven gusts. HRDPS: no gust field. Plan smooth, not puffy.

Sea state — benign

~0.2 m / 0.7 ft short-period wind chop (NWS seas ≤1-2 ft). SST ~21°C / 69°F. Flat water — the limiter is wind, not waves.

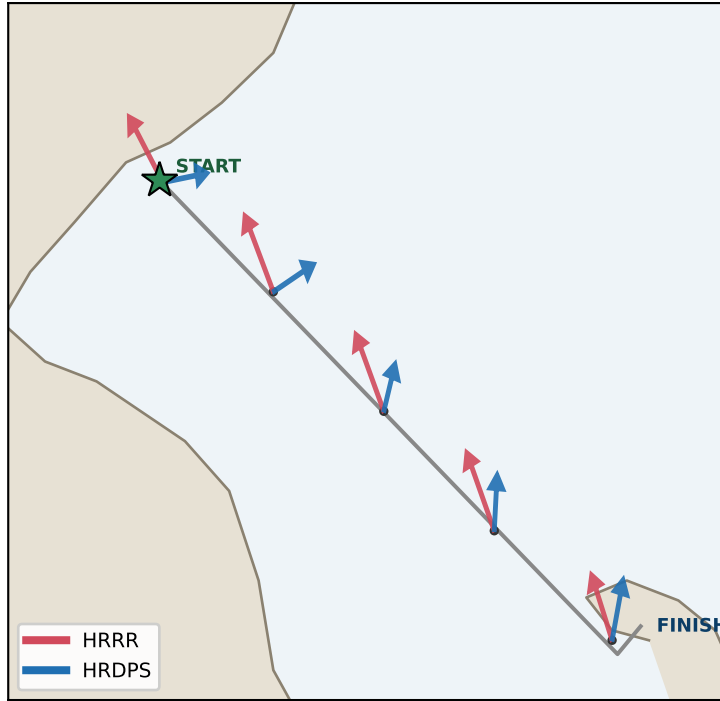
Temperature — mild & near-steady

~18°C / 64-65°F all night on the water (sea-moderated: ~1°F drop offshore, up to ~10°F near the P-town shore). Warm water under cooler air → damp; reinforces the fog watch. Layers for damp, not cold.

Tactics, Strategy & Timing

Marblehead → Provincetown · Fri 2026-07-17 overnight

Where the models differ — mean-wind arrows (length $\propto$ speed)



HRRR vs HRDPS — mean wind by point

Start (Tinkers Rk)

HRRR 145°/7kt · HRDPS 261°/5kt → Δ dir 116°

Off Nahant

HRRR 153°/8kt · HRDPS 243°/5kt → Δ dir 90°

Mid Mass Bay

HRRR 154°/8kt · HRDPS 198°/3kt → Δ dir 44°

Stellwagen edge

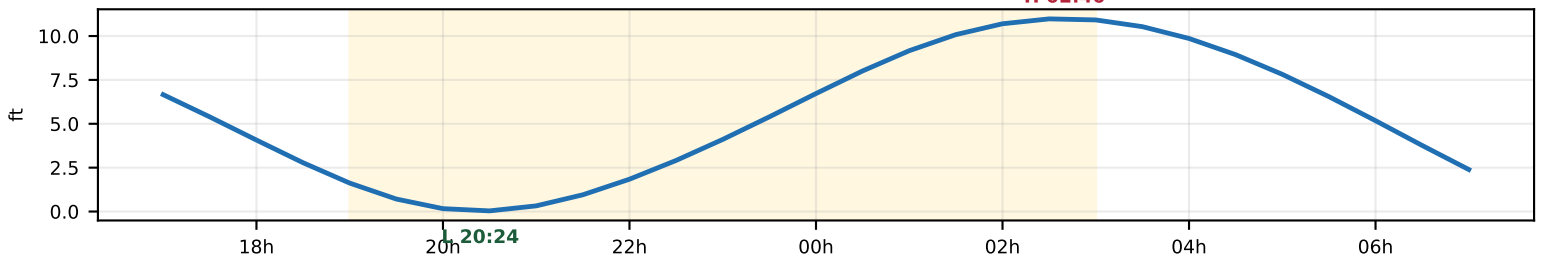
HRRR 155°/8kt · HRDPS 184°/4kt → Δ dir 29°

Race Pt / P-town

HRRR 157°/6kt · HRDPS 194°/5kt → Δ dir 37°

Biggest split at the START (near-opposite: a dying SE sea-breeze vs a NW land-breeze). They converge to a common S/SW by Race Point, HRDPS lighter throughout. → The first leg is the uncertain part — sail the actual breeze; for the rest of the night both models agree.

Tide at Boston (proxy) — start $\approx$ low slack, finish $\approx$ high slack



Tactics & strategy

① Start (19:00-21:00) — models disagree, so read the water

HRRR: SE ~10 kt (reach/close-hauled). HRDPS: NW land-breeze ~7 kt (offshore), dying to calm ~01:00. Opposite angles — don't pre-commit. Sail the actual dock breeze; whatever it is it's LIGHT (≤ 8 kt). Get clear air off the line, then work OFFSHORE — the Marblehead shore is the light trap.

② The crossing (21:00-01:00) — reach in the filling southerly

Both models fill S/SW 8-12 kt offshore — a starboard reach that frees as the wind veers right. Pressure lives offshore/south (mid-bay ~10 kt vs shore 2-6). Commit south; stay on the pressure side, don't over-stand north toward Cape Ann.

③ ⚠️ LNG Northeast Gateway — hard avoid

Two STL buoys at ~42.40 N (–70.60/–70.59) sit just N of the direct line. NO ENTRY within 500 m, no anchoring within ~1 nm. The rhumb passes ~5 nm south — staying on/south of the line is both faster (more breeze) and legal.

④ Race Point & finish (01:00-03:00) — the tide gate

S/SSW 8-11 kt, gusts to 15 into Cape Cod Bay. Flood builds to Boston high 02:45 $\approx$ finish. Round Race Point BEFORE ~02:45 to carry the flood in; later boats meet the building ebb. Finish current itself is minor (near high slack).

⑤ Wind shear — expect shifty, veered puffs (full explainer + trim, p.3)

Light SE/S surface sits under a W-WNW 13-16 kt flow at 925/850 hPa with a low-level inversion (decoupled night). Puffs mix a little of that down — shifter and veered toward W/NW vs the mean, but capped ~16 kt. Toward dawn the layer mixes and the surface veers/builds to the gradient. Don't bank on one wind angle; sail conservatively through puffs, keep the boat in pressure.

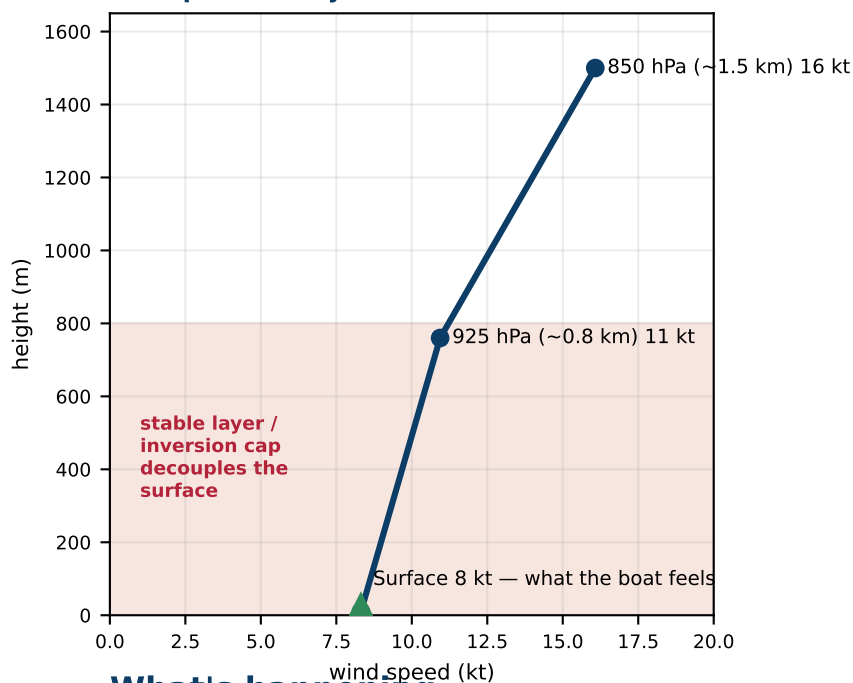
⑥ Night & hazards

Dark — thin crescent sets early, so nav by instruments, watches + lights set. Fog watch: July S-flow over ~20°C water; NWS not flagging Fri night but recheck AM. Seas ≤ 2 ft, no SCA expected.

Wind Shear Tonight — What It Means for Your Race

A light surface wind sits under a stronger, differently-directed flow aloft — the defining feature of this overnight.

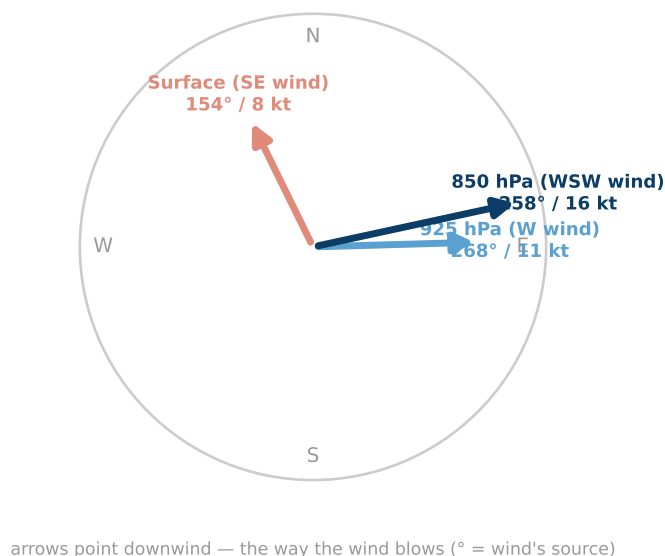
Speed nearly doubles in the first km



What's happening

After sunset the sea-cooled surface layer decouples from the wind above — HRRR shows a low-level inversion (925 hPa is +1.2°C WARMER than the surface, a stable cap). Below it the course feels a light, thermal breeze (dying SE sea-breeze / land-breeze); above it the true gradient wind, W-WNW 13–16 kt, flows almost unimpeded. The two are ~100° apart and the speed nearly doubles in the first km. A puff is a brief moment when some of that faster, veered air mixes down; toward dawn the whole layer re-mixes and the surface swings toward the gradient.

Wind veers ~100° clockwise with height



Tactical consequences for sailing

• Puffs come in veered AND stronger

Not just more pressure — the puff arrives from further right (toward W/NW). Upwind: starboard lifts, port headers; reaching it frees then heads. Anticipate the shift, don't just trim for speed.

• Play the shifts — don't commit a side

Light and shifty rewards the boat that tacks on headers and keeps the bow in pressure, not a big side bet. The models even disagree on direction, so trust what you see over any single number.

• Trim for twist

The masthead sees a stronger, veered wind than the deck. Add twist — ease the upper leech (traveller down / vang off a touch upwind; more headsail twist). A closed leech stalls or backwinds up top.

• Masthead ≠ deck

Your wind vane and the water/telltales will disagree — that's the shear, not a fault. Steer to the jib telltales + boat feel; read the masthead fly to see the next puff/veer coming.

• Expect a dawn build & veer

As mixing returns near sunrise the gradient reaches the surface: the breeze should veer toward W/SW and build. Set up for the new wind before it fills — it comes from the gradient (offshore/right) side.

• Hunt where it mixes down

The fast air aloft is a reservoir; darker, textured water marks where it's reaching the surface. Sail toward the pressure — but it's capped ~16 kt, so gains are about staying IN pressure, not surviving gusts.